

A Symmetric Six-Point Counterexample to Concavity of Determinantal Entropy

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Abstract

Lyons asked whether the Shannon entropy of a finite determinantal probability measure is concave in its positive-contraction kernel. We give a counterexample on six labelled points. Its midpoint is a real symmetric matrix obtained by moving a rank-three projection into the interior, and its two endpoints differ from the midpoint by opposite purely imaginary Hermitian matrices. Symmetry reduces the 64 complete-event probabilities to eight pairs. A short exact calculation proves that the endpoint laws agree and that their entropy is strictly larger than the entropy at the midpoint.

1 The example

Let $E = [6]$. For a Hermitian matrix L with $0 \preceq L \preceq I$, write X_L for the determinantal process characterized by

$$\Pr(T \subseteq X_L) = \det L[T] \quad (T \subseteq E),$$

and write

$$p_A(L) = \Pr(X_L = A), \quad H(L) = - \sum_{A \subseteq E} p_A(L) \log p_A(L).$$

The conjecture that H is concave in L was posed by Lyons [1] and restated as Conjecture 2.6 in his ICM survey [2].

Theorem 1. *There are strict positive contractions Q_+ and Q_- whose midpoint K satisfies*

$$p_A(Q_+) = p_A(Q_-) \quad (A \subseteq E)$$

and

$$\frac{1}{40\,000\,000} < H(Q_+) - H(K) < \frac{1}{39\,000\,000}. \quad (1)$$

Consequently,

$$H(K) < \frac{H(Q_-) + H(Q_+)}{2}.$$

Thus determinantal entropy is not concave on the set of complex Hermitian positive contractions.

We now give the matrices. Consider the two 3×3 matrices

$$R = \frac{1}{3} \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix},$$

and put

$$P = \frac{1}{2} \begin{pmatrix} I_3 & R \\ R & I_3 \end{pmatrix}, \quad B = \begin{pmatrix} S & 0 \\ 0 & -S \end{pmatrix}. \quad (2)$$

Finally, set

$$\varepsilon = \frac{1}{475}, \quad t = \frac{1}{2300}, \quad K = \varepsilon I_6 + (1 - 2\varepsilon)P, \quad Q_{\pm} = K \pm itB. \quad (3)$$

Numerically, the entropy increase in (1) is $2.517182345129681 \dots \times 10^{-8}$. For scale,

$$H(K) = 2.877726131217012 \dots, \quad H(Q_+) = 2.877726156388836 \dots;$$

the increase is about 8.75×10^{-9} times the entropy at the midpoint.

2 The structural checks

Everything needed to see that the matrices in (3) are legal is contained in the following identities.

Lemma 1. *The matrices in (2) satisfy*

$$R^T = R, \quad R^2 = I_3, \quad S^T = -S, \quad RS = -SR.$$

Hence P is a rank-three orthogonal projection, $B^T = -B$, and $PB = BP$. Furthermore, $\|iB\| = \sqrt{3}$.

Proof. The four displayed identities follow by direct multiplication. Block multiplication then gives $P^T = P = P^2$, $B^T = -B$, and $PB = BP$. The characteristic polynomial of iS is $\lambda(\lambda^2 - 3)$, which gives the norm assertion. $\square$

The eigenvalues of K are ε and $1 - \varepsilon$, each with multiplicity three. Since

$$t\|iB\| = t\sqrt{3} < \varepsilon,$$

the usual eigenvalue perturbation bound places every eigenvalue of $Q_{\pm}$ strictly between 0 and 1. Thus $Q_{\pm}$ are strict positive contractions.

For later use, recall the complete-event formula

$$p_A(L) = (-1)^{|E \setminus A|} \det(L - I_{E \setminus A}), \quad (4)$$

where I_C denotes the diagonal projection onto the coordinates in C . It follows immediately from inclusion–exclusion applied to the defining inclusion probabilities.

Because $Q_- = \overline{Q_+}$, the two determinants in (4) are complex conjugates. They are real because their matrices are Hermitian. Therefore

$$p_A(Q_+) = p_A(Q_-) \quad (A \subseteq E). \quad (5)$$

This proves the equality of the endpoint laws without numerical computation.

3 The entropy comparison

It remains to establish the strict inequality in (1). The following table contains the entire calculation. Put

$$u = \varepsilon(1 - \varepsilon).$$

Coordinate permutations preserving P and sending B to either B or $-B$, together with complementation, divide the 64 subsets into the eight classes shown below. For example, the coordinate symmetries are generated by $(1\ 2\ 3)(4\ 6\ 5)$, $(2\ 3)(4\ 6)$, and $(1\ 4)(2\ 5)(3\ 6)$. Conjugation by $\text{diag}(I_3, -I_3)$ sends Q_- to $I - Q_+$; the standard complementation rule for DPPs and (5)

therefore identify a subset with its complement. Substitution into (4) gives the two right-hand columns.

Here is one row in full. Since K has eigenvalues ε and $1 - \varepsilon$, each three times,

$$p_{\emptyset}(K) = \det(I - K) = \varepsilon^3(1 - \varepsilon)^3 = u^3.$$

The commutation $PB = BP$ makes both the range and the kernel of P invariant under iB . Under the identifications $x \mapsto (x, Rx)$ and $x \mapsto (x, -Rx)$, the restriction of iB to either three-dimensional subspace is represented by iS , whose eigenvalues are $0, \sqrt{3}, -\sqrt{3}$. Therefore

$$\begin{aligned} p_{\emptyset}(Q_+) &= \det(I - Q_+) \\ &= \varepsilon(1 - \varepsilon)(\varepsilon^2 - 3t^2)((1 - \varepsilon)^2 - 3t^2) \\ &= u^3 + 3t^2u(3t^2 + 2u - 1). \end{aligned}$$

This is exactly the first row of the table. The other seven rows follow from the same determinant formula; the checker verifies every representative and every multiplicity.

representative A	multiplicity	$p_A(K)$	$p_A(Q_+) - p_A(K)$
$\emptyset$	2	u^3	$3t^2u(3t^2 + 2u - 1)$
$\{1\}$	12	$u^2(1 - 2u)/2$	$-t^2(18t^2u - 3t^2 + 12u^2 - 6u + 1)/2$
$\{1, 2\}$	12	$u(1 - 2u)^2/4$	$t^2(18t^2u - 6t^2 + 12u^2 - 8u + 1)/2$
$\{1, 2, 3\}$	2	$(1 - 2u)^3/8$	$3t^2(1 - 2u)(3t^2 + 2u - 1)/2$
$\{1, 4\}$	12	$u(36u^2 - 20u + 5)/36$	$t^2(54t^2u - 12t^2 + 36u^2 - 10u + 1)/6$
$\{2, 4\}$	6	$u(9u^2 - 8u + 2)/9$	$t^2(27t^2u - 6t^2 + 18u^2 - 11u + 2)/3$
$\{1, 2, 4\}$	12	$(1 - 2u)(9u^2 - 4u + 1)/18$	$-t^2(54t^2u - 15t^2 + 36u^2 - 20u + 2)/6$
$\{1, 3, 4\}$	6	$(1 - 2u)(36u^2 - 4u + 1)/72$	$-t^2(54t^2u - 15t^2 + 36u^2 - 8u - 1)/6$

For completeness, we record how the numerical sign is certified without floating-point assumptions. If a positive rational number is written as $x = 2^k r$, where $1 \leq r < 2$, and $y = (r - 1)/(r + 1)$, then for every positive integer m ,

$$2 \sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} \leq \log r \leq 2 \sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} + \frac{2y^{2m+1}}{(2m+1)(1-y^2)}. \quad (6)$$

The same formula at $r = 2$ gives rational lower and upper bounds for $\log 2$. Thus (6) gives directed rational bounds for the logarithm of every entry in the table.

For a row of multiplicity m , with center probability p and endpoint change d , its contribution to the entropy difference is $m[-(p+d) \log(p+d) + p \log p]$. Substituting $\varepsilon = 1/475$ and $t = 1/2300$, summing the eight rows, and taking 50 terms in (6) gives, by rational arithmetic,

$$\frac{1}{40\,000\,000} < H(Q_+) - H(K) < \frac{1}{39\,000\,000}.$$

This is (1). Together with (5), it completes the proof of Theorem 1.

The accompanying checker starts from the four displayed matrices and reproduces every event probability, the eight-row grouping, endpoint feasibility, equality of the endpoint laws, and the two directed entropy bounds using only exact rational arithmetic and the Python standard library.

4 Why the direction works

The calculation above also has a simple conceptual explanation. If L is real symmetric and B is real skew-symmetric, transposition in (4) shows that

$$p_A(L + isB) = p_A(L - isB).$$

Every complete-event probability is therefore an even function of s , so its first derivative at $s = 0$ vanishes. Twice differentiating entropy gives

$$\left. \frac{d^2}{ds^2} H(L + isB) \right|_{s=0} = - \sum_{A \subseteq E} \left. \frac{d^2}{ds^2} p_A(L + isB) \right|_{s=0} \log p_A(L). \quad (7)$$

The normally negative term involving the squares of the first derivatives has disappeared. Formula (7) says that the sign is decided entirely by the second-order motion of the event probabilities. In information-geometric language, this is the second fundamental form of the probability map paired with the entropy gradient.

The two displayed matrices are one member of a simple algebraic family. If R is any real symmetric matrix with $R^2 = I$, and S is any real skew-symmetric matrix satisfying $RS = -SR$, then the block formulas

$$P = \frac{1}{2} \begin{pmatrix} I & R \\ R & I \end{pmatrix}, \quad B = \begin{pmatrix} S & 0 \\ 0 & -S \end{pmatrix}$$

automatically give an orthogonal projection P and a commuting skew direction B . The particular rational pair above was selected within this family because its event probabilities have only eight classes and its second-order motion raises entropy.

This family was found by searching for first-order degeneracy near the boundary of the set of positive contractions. That search motivates the construction but is not needed to verify it. Once R and S are given, the proof consists of the matrix identities in Lemma 1, the event formula (4), and the eight-row entropy calculation.

The midpoint in the example is real symmetric, whereas the direction iB is purely imaginary Hermitian. The example therefore disproves the conjecture with complex Hermitian kernels. It does not settle the separate restriction to real symmetric kernels, nor does it establish that six is the smallest possible dimension.

The construction was found with Qwen Euler agent assistance. Numerical search was used to locate the pattern; all statements in this note and in the checker are verified by exact arithmetic once the matrices R and S are given. To the best of our knowledge, the cited conjecture had no previously published proof or counterexample.

References

- [1] Russell Lyons. Determinantal probability measures. *Publications Mathématiques de l’IHÉS*, 98:167–212, 2003.
- [2] Russell Lyons. Determinantal probability: Basic properties and conjectures. In *Proceedings of the International Congress of Mathematicians 2014*, volume IV, pages 137–161, 2014.